Class 21: Clusters on Demand

k-means always finds groups. When should you believe them?

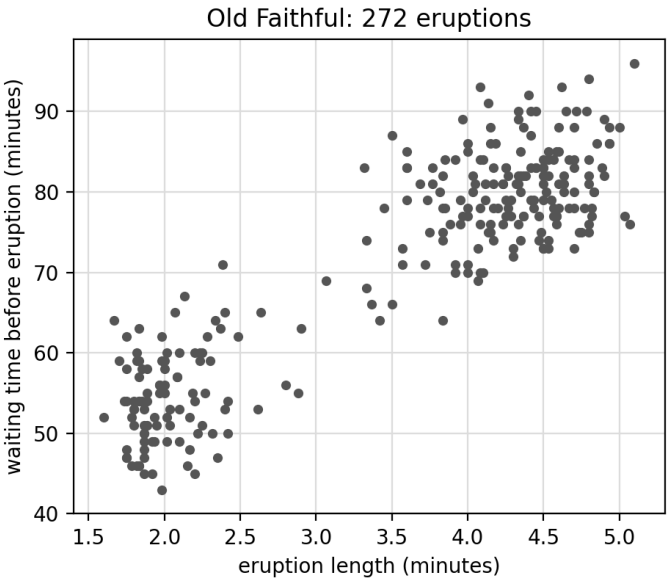
Team Members: _____ Date: _____

Clustering sorts points into groups using only how close they are to each other – no labels, no right answers given. The most common method is **k-means**. You choose k, the number of groups, and the algorithm does the rest:

Start. Pick k rows at random to be the starting centers. (The `seed` decides which rows.)
Step A. Every point joins the center it is closest to.
Step B. Every center moves to the average of its points.
Repeat A and B until no point changes groups.

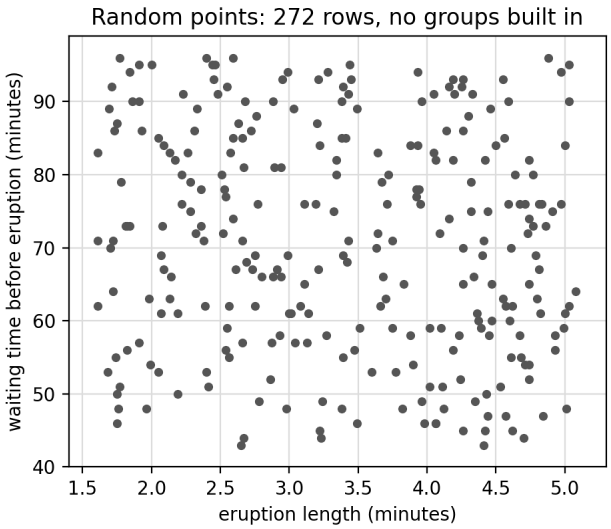
Both columns are converted to standard units first, so a minute of waiting doesn't count more than a minute of eruption just because waiting times are bigger numbers.

Note: Lab 10 uses **k-nearest neighbors**, a different method. There, k counts neighbors used to make a prediction. Here, k counts groups.



Q1 – Before running any code. How many groups do you see? Draw the boundary between them on the plot. Describe each group in a few words.

Q2. Run `kmeans(faithful, 2, seed=0)`. Does its boundary match yours? How many eruptions are in each cluster?



Q3 – Predict first. These 272 points were generated at random, spread evenly over the same ranges. What will `kmeans(noise, 2, seed=0)` do? Circle one, then sketch your guess on the plot.

(a) give an error or refuse
(b) put nearly every point in one cluster
(c) split the points into two clusters of about equal size
(d) something else: _____

Q4. Run k = 2 on both datasets with seed 0, then with seed 2. Fill in the table. For "boundary," write vertical, horizontal, or diagonal.

	seed 0: sizes	**seed 0: boundary**	**seed 2: sizes**	**seed 2: boundary**
Old Faithful				
Random points				

Which dataset's answer depended on the seed? _____

Q5. Now ask for k = 3 on Old Faithful, with seed 0 and then seed 6. Record the cluster sizes. Did k-means complain? Which, if any, of the three clusters do you believe?

seed 0: _____ seed 6: _____

Q6. k-means always returns exactly k groups. Name two pieces of evidence you would want before believing a cluster is real.